Course Outline

Sem		Course Name	Th	Tut	Pra	Hrs	Cr
Sem I	HSC	Humanities – 1	2	0	0	2	2
	BSC	Mathematics – 1	3	1	0	4	4
	BSC	Physics	3	0	0	3	3
	BSC	Physics – Laboratory	0	0	2	2	1
	BSC	Environment Science	2	0	0	2	2
	ESC	Workshop Practices	0	0	2	2	1
	BSC	Biology for Engineers	2	0	0	2	2
	ESC	Computer Programming -I	1	0	0	1	1
	ESC	Computer Programming -I Laboratory	0	0	2	2	1
	HSC	Universal Human Values	1	0	0	1	1
	HSC	Indian Knowledge System	2	0	0	2	2
			16	**1**	**5**	**23**	**20**
Sem II	HSC	Humanities – 2	2	0	0	2	2
	BSC	Mathematics – 2	3	1	0	4	4
	BSC	Chemistry	3	0	0	3	3
	BSC	Chemistry – Laboratory	0	0	2	2	1
	ESC	Introduction to Electrical Engineering	3	0	0	3	3
	ESC	Introduction to Electrical Engineering-Laboratory	0	0	2	2	1
	ESC	Engineering Graphics	0	0	4	4	2
	HSC	Yoga, Health & Hygiene OR NCC/NSS	0	0	2	2	1
	HSC	Organizational Behaviour	1	0	0	1	1
	ESC	Computer Programming -II	1	0	0	1	1
	ESC	Computer Programming -II Laboratory	0	0	2	2	1
			13	**1**	**12**	**26**	**20**
Sem III	Pro	Civic and Social Service Internship	0	0	0	0	1
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	PC	Digital Circuits	3	0	0	3	3
	PC	Digital Circuits Laboratory	0	0	2	2	1
	PC	Electronics Devices and Circuits	3	0	0	3	3
	PC	Electronics Devices and Circuits Laboratory	0	0	2	2	1
	PC	Networks and Systems	3	1	0	4	4
	BSC	Math-3	3	1	0	4	4
			15	**2**	**4**	**21**	**20**
Sem IV	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 – Laboratory (Department Specific)	0	0	2	2	1
	OE	Open Elective 1 (From Other School)	3	0	0	3	3
	PC	Analog Electronics	3	0	0	3	3
	PC	Analog Electronics Laboratory	0	0	2	2	1
	PC	Analog Communication	3	0	0	3	3
	PC	Analog Communication Laboratory	0	0	2	2	1
	PC	Electromagnetics and Transmission Lines	3	1	0	4	4
	PC	Digital Signal Processing	3	0	0	3	3
	PC	Digital Signal Processing Laboratory	0	0	2	2	1
			17	**1**	**8**	**26**	**22**
Sem V	OE	Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3

Semester 3

24ICxxxT					Digital Circuits					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	0	0	100

COURSE OBJECTIVES

- To learn the fundamentals of digital logic circuits and digital number systems.
- To analyze and design digital combinational-logic and sequential-logic circuits.
- To impart to student the concepts of sequential circuits, enabling them to analyze sequential systems in terms of state machines.

UNIT-1: NUMBER SYSTEMS AND LOGIC GATES **9 Hrs.**

Digital systems; positional number system; binary, octal and hexadecimal representation of integer/fixed-point numbers; Methods of base conversions; complements; Representation of signed numbers; signed/unsigned binary, octal and hexadecimal arithmetic; Binary coded decimal codes; Gray codes; Excess-3 codes, Logic Gates, and Truth Table; Error detection and correction codes - parity check codes and Hamming codes.

UNIT-2: BOOLEAN ALGEBRA AND COMBINATIONAL CIRCUITS **13 Hrs.**

Boolean Algebra & Simplification of Boolean Algebra: Basic postulates and fundamental theorems of Boolean algebra; Representation of logic functions: canonical and standard forms; Simplification of Boolean functions: K-map and Quine-McCluskey tabular methods.

Definition of combinational circuits; design procedure; Decoders; encoders; multiplexers; demultiplexers and their applications; Parity generator and checker circuits; comparators; Arithmetic modules- ripple carry adder, subtractor, carry look ahead adder; basic multiplier, BCD Adder; Shannon expansion theorem.

UNIT-3: SEQUENTIAL CIRCUITS **14 Hrs**.

Definition of sequential circuits; Latches and flip-flops: SR-latch; D-latch; D flip-flop; JK flip-flop; T flip-flop.
Synchronous sequential circuit; Moore and Mealy Finite State Machines (FSM): state diagram, state table, state assignment; Analysis and implementation of FSM using D flip-flops, JK flip-flops and T flip-flops; Implementation of FSM using Decoders, OR gates, Multiplexors and ROM.
Various Registers and counters: asynchronous counters, synchronous counters; ring counter, Johnson counter.

UNIT 4: INTRODUCTION TO MEMORY AND PROGRAMMABLE LOGIC CIRCUITS **6 Hrs**.

Introduction, Random access memory, read only memory, programmable logic array, programmable array logic, sequential programmable devices, FPGA.

Max. Hrs.: 42

COURSE OUTCOMES

On completion of the course, the student will be able to
CO1: Study the basics of number systems, Boolean algebra and standard digital logic gates.
CO2: Understand the knowledge of Boolean Algebra and simplification of Boolean expressions to deduce optimal digital networks.
CO3: Apply the knowledge to synthesis combinational circuits.
CO4: Analyse the operation and logic circuits of basic storage elements FFs.
CO5: Implement SSI and MSI digital networks given a state diagram based on Mealy and Moore configurations.
CO6: Design and examine the SSI, MSI and Programmable combinational networks.

TEXT/REFERENCE BOOKS

1. Digital Design - M. Morris Mano and Michael Ciletti, 5 th edition, Pearson.
2. Switching Theory and Logic Design - A. Anand Kumar, PHI, 2nd Edition.
3. Fundamentals of Digital Logic with Verilog Design - Stephen Brown and Zvonko Vranesic Tata McGraw Hill.
4. Switching and Finite Automata Theory - Zvi Kohavi & Niraj K. Jha, 3rd Edition, Cambridge.
5. Fundamentals of Logic Design - Charles H. Roth, Cengage LEanring, 5th, Edition.
6. Introduction to Switching Theory and Logic Design – Fredriac J. Hill, Gerald R. Peterson, 3rd Ed, John Wiley & Sons Inc.
7. Digital Fundamentals – A Systems Approach – Thomas L. Floyd, Pearson.

20ECExxxP					Digital Circuits Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	25	25	50

COURSE OBJECTIVES:

- To learn the fundamentals of digital logic design and digital circuits.
- To analyze and design digital combinational-logic and sequential-logic circuits.
- To implement digital combinational-logic and sequential-logic circuits.

List of Experiments

Perform the below mentioned experiments using IC trainer kit and Digital ICs

1. Study of Different types of logic gates and implement them using NAND and NOR gates.
2. Design a simple combinational circuit with four variables and obtain minimal expression and verify the truth table.
3. Study of Half and full adder circuits.
4. Study of Binary and BCD parallel adder circuits.
5. Study of Decoder and Encoder circuits.
6. Study of Multiplexer and De-multiplexer circuits.
7. Study of Different types of flip flops.
8. Study of Asynchronous and synchronous binary counters.
9. Study of Various types of registers.
10. Design a BCD to 7-segment display decoder circuit and interface with a 7-sgment display.

COURSE OUTCOMES

On completion of the course, the student will be able to:

CO1 - Remember digital logic circuit functionalities

CO2 - Understand the design methodologies to utilize them practically.

CO3 - Apply design methods to realize digital circuits practically.

CO4 - Analyze functionalities of digital circuits.

CO5 - Evaluate and report performance of digital circuits.

CO6 - Design and Test elementary digital combinational-logic and sequential-logic circuits.

TEXT/REFERENCE BOOKS

1. M. Morris Mano and Michael Ciletti, "Digital Design: With a Introduction to the Verilog HDL", 5th edition,Pearson.

<Course Code>					Electronics Devices and Circuits					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES –

- To offer profound understanding of the fundamentals of semiconductor devices.
- To impart comprehensive knowledge of device technology.
- To provide a foundation in the physics and technology of semiconductor devices in order to address the challenges of modern electronic devices.

UNIT I: FUNDAMENTALS OF SEMICONDUCTOR DEVICES	**14 Hrs.**
Evolution and scope of semiconductor devices and technology, Drift velocity, Einstein relation, conductivity, resistivity, Drift current, diffusion current density, Types of semiconductors, Energy bands formation in semiconductors, Fermi-Dirac distribution, Fermi level in intrinsic/extrinsic semiconductors and dependency with temperature and doping, Carrier concentrations at equilibrium, Temperature-dependence on carrier concentrations, Mobility, conductivity, Carrier generation and recombination.	
UNIT II: PN JUNCTION DIODE AND IT'S APPLICATIONS	**8 Hrs.**
Junctions: Metal/Semiconductor (Schottky barrier height, rectifying and ohmic contacts), Mathematical derivation of the potential barrier, depletion width, and I-V characteristics, Junction break down (Zener and avalanche effects), Zener diode, Clipper, Clamper, Voltage multiplier, voltage regulator, Passive filters.	
UNIT III: BIPOLAR JUNCTION TRANSISTOR	**6 Hrs.**
Working of a BJT, transistor biasing, different transient circuit configuration (CB, CE and CC), static characteristic for BJT, transistor as switch, amplifier.	
UNIT IV: MOS TRANSISTOR	**12 Hrs.**
Introduction to MOSFET: structure, types and its operation, MOS Capacitor, Threshold voltage, Mathematical derivation for I-V, Transfer and O/P characteristics, Second order effects, Scaling of MOSFET, Short channel effects: DIBL, GIDL, Hot carrier injection, surface scattering impact ionization.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Define the basic concepts of semiconductor technology, and the importance of semiconductors in modern electronic devices.
CO2	:	Understand the charge transport mechanism, operation and working principle of PN junction diode.
CO3	:	Apply the various techniques in semiconductor devices and MOS Transistors.
CO4	:	Analyze the various issues of BJT, MOS devices with size scaling and also analyze the theoretical models of MOS transistors.
CO5	:	Evaluate the performance of semiconductor devices.
CO6	:	Build the device analysis framework to improve the semiconductor devices for specific desired real-life applications with considering issues and constraints at research and industry levels.

TEXT/REFERENCE BOOKS

1. B.G. Streetman & S. Banerjee, "Solid State Electronic Devices", PHI.
2. Donald A. Neamen, "Semiconductor Physics and Devices", 4th Edition, McGraw Hill Education.
3. Boylestad and Nashlesky, "Electronic Devices and Circuit Theory", PHI
4. N.N. Bhargava, S.C. Gupta, and D.C. Kulshreshtha, "Basic Electronics And Linear Circuits", McGraw Hill Education.

<Course Code>					Electronics Devices and Circuits Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To understand the characteristics of PN junction diodes and their applications.
- To observe properties of MOSFET.
- To illustrate the MOSFET application in different real life circuits.

LIST OF EXPERIMENTS	
1	To study the simulation tool and its features for analog circuit simulation.
2	To study the VI characteristic of silicon and germanium diodes.
3	To study reverse characteristics of zener diode.
4	To design diode Clipper and clamper Circuits
5	To design Voltage doubler circuit
6	To design Voltage tripler and quadrupler circuit
7	To design low and high pass filter.
8	To study BJT as switch.
9	To To study common emitter amplifier.
10	To study transfer and drain characteristics of MOSFET.
11	To study and design MOSFET as a switch.
12	To study and design common source amplifier using MOSFET.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Study the fundamentals of electronic components.
CO2	:	Understand the working principle of semiconductor devices.
CO3	:	Apply the different techniques for building real time circuits.
CO4	:	Analyze the behaviour of semiconductor devices, PN junction diode and MOSFET
CO5	:	Evaluate different circuit for different device parameters.
CO6	:	Build PN diode and MOSFET based sub-system.

<Course Code>					Networks and Systems					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand classification of signals and systems
- To learn applications of mathematical tools like Laplace Transform and Fourier Transform in circuit analysis.
- To understand the importance of two port networks.

UNIT I: CONTINUOUS TIME SIGNALS, CIRCUITS AND SYSTEMS	**10 Hrs.**
Signal – Definition, Examples, Classifications, Continuous time signals: Energy, power, Periodicity, Signal operations, Elementary Signals, System - Definition, Classification, Continuous time LTI systems models in time domain: Linear Constant Co-efficient Differential Equations, Convolution integral and unit impulse response, Simple circuits using R, L, C and controlled sources as LTI systems and their time domain descriptions, Interconnections of Systems, Stability and Causality.	
UNIT II: LAPLACE TRANSFORM APPLICATIONS TO SYSTEM AND CIRCUIT ANALYSIS	**11 Hrs.**
Review of Laplace transform: Definition, Elementary pairs, Basic properties, Region of convergence (ROC), Inverse Laplace transform, Application to LTI system analysis, System transfer function, poles and zeros: stability and causality, Laplace transform for solution of linear constant co-efficient differential equation, Application to transient analysis of R,L,C circuits with relaxed and non-relaxed conditions, Application to waveform analysis.	
UNIT III: FOURIER TRANSFORM APPLICATIONS TO SYSTEM AND CIRCUIT ANALYSIS	**10 Hrs.**
Review of Fourier series and Fourier transform, Signal spectra, Frequency analysis of LTI systems, Application to sinusoidal steady-state analysis of R,L,C circuits with controlled sources, Frequency response plots and concept of filters.	
UNIT IV: MULTI PORT CIRCUITS ANALYSIS	**11 Hrs.**
Single port and two port circuits with R, L, C and Controlled sources, Source transformations, Driving point functions, Transformed circuits and analysis using Super-position, Thevenin and Norton‟s theorem, Two port network analysis using z, y and h parameters, Transmission line as two port network and ABCD parameters, series and parallel connections of two port circuits and their parameters, Applications in linear electronic circuit and system analysis.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember properties of continuous time signals and systems in Time domain, s-domain and frequency domain.
CO2	:	Understand concepts and complexity of circuit and system analysis techniques.
CO3	:	Apply concepts to analyze LTI systems and linear circuits.
CO4	:	Analyze LTI systems and linear circuits using mathematical tools.
CO5	:	Evaluate signal response in linear circuit using different techniques.
CO6	:	Create mathematical models for linear circuits and other LTI systems.

TEXT/REFERENCE BOOKS

1. Signals & Systems, Alan V.Oppenheim, Alan S.Willsky with S. Hamid Nawab, Pearson Education.
2. Network Analysis, M E Van Valkenburg, PHI.
3. Circuits, Systems and Signal Processing, A Tutorials Approach, Suhash Chandra Dutta Roy, Springer.
4. Circuits and Systems, K. M. Soni, S. K. Kataria & Sons.